MIZAN

Arabic Morphological Pattern Analyzer

User Guide · Version 2.0

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Moataz Tools · 2026

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1 Please Cite This Tool

⚠ If you use MIZAN in academic research, you are requested to cite the associated publication. Citing ensures proper attribution and supports continued work in Arabic computational linguistics.

Al-Said, A. B. (2026). Mizan: An Innovative Tool for Building Arabic Morphological Pattern Dictionaries and Enhancing Lexical Resource Development. ACM Transactions on Asian and Low-Resource Language Information Processing, 25(2), Article 11.

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```

2 Introduction

MIZAN is a computational tool for Arabic morphological pattern analysis at scale. Given a user-defined template (وزن), it identifies every word conforming to that template across a large Arabic text corpus and produces structured frequency lists suitable for lexicographic research, NLP development, and computational linguistics.

Unlike full-parsing tools such as BAMA, SAMA, MADAMIRA, Farasa, and CamelMorph — which assign complete morphological analyses to individual tokens — MIZAN focuses on pattern-based extraction at corpus scale, processing hundreds of millions of words to find all instances of a given morphological class.

2.1 Key Capabilities

- Scale — processes corpora exceeding 500 million words.
- Pattern extraction — matches any Arabic morphological template using ف–ع–ل notation, with or without diacritics.
- Normalization — strips tashkeel, normalises alef variants, and applies prefix/suffix rules automatically.
- Recursive search — traverses all subfolders without additional configuration.
- Structured output — tab-separated files with five columns: Rank, Word, Count, Pct%, Cumulative%.
- KWIC contexts — generates Keyword in Context files for qualitative validation of results.
- Concordance viewer — click any word in the Results table to open a full corpus concordance in a dedicated dialog.
- Corpus statistics — actual token and type counts, Type-Token Ratio, per-file breakdown, and cross-pattern overlap analysis.
- Session log — complete timestamped processing log saved automatically with every session.

2.2 System Requirements

Component	Requirement
Operating System	Windows 10/11 · Linux · macOS
Python Version	3.10 or higher
Required Library	PyQt6 (pip install PyQt6)
Optional Library	psutil (pip install psutil) — adaptive memory management
Recommended RAM	4 GB minimum · 8 GB for corpora above 100 million words
Corpus Format	Plain text (.txt) UTF-8 or XML (.xml)

2.3 Installation and Launch

```
pip install PyQt6 psutil
python mizan_v20.py
```

⚠ Windows: always launch from cmd.exe. Launching from IDLE or Spyder may cause the GUI to fail silently.

3 Interface Overview

The interface is divided into two panels separated by a resizable splitter.

3.1 Left Panel — Configuration

The left panel contains all input controls, organized into five groups.

Corpus & Output

Field	Description
Corpus Folder	Root folder of the text corpus. All subfolders are searched recursively.
Output Folder	Destination for the session folder. MIZAN creates a timestamped subfolder automatically.
Stopwords (optional)	Path to a plain-text file of additional stopwords, one per line. Leave blank to use built-in Arabic stopwords.

Format & Encoding

Option	Description
Plain Text (.txt)	Standard UTF-8 text files. Recommended for most corpora.
XML (.xml)	Structured corpora such as Arabic Wikipedia dumps or CAMeL-formatted files.
UTF-8	Standard Unicode — recommended.
CP1256	Windows Arabic legacy encoding.
UTF-16	Unicode with byte-order mark — common in Microsoft Word exports.
ISO-8859-6	ISO Latin/Arabic standard encoding.

Morphological Patterns

Patterns use the classical Arabic root notation where ع, ف, and ل represent the three root letters, and all other characters are fixed structural letters. For example:

Pattern	Structure
مَفْعُول	Passive participle — م is fixed; ل, ع, ف are root slots
اِسْتِفْعَال	Verbal noun Form X — اِسْتِ and ال are fixed; ل, ع, ف are root slots
فَعَّال	Intensive adjective — ل, ع, ف are root slots; the shadda on ع is fixed

Patterns can be added in two ways:

- Type any Arabic pattern in the input field and press ＋ Add or Enter.
- Click any button in the Quick Examples grid to add a pre-defined pattern instantly.

ℹ The input field accepts any valid Arabic pattern, including patterns not listed in the Quick Examples. Enter custom patterns such as مستفعلات or فاعولة directly.

Analysis Options

Option	Behaviour
Strip diacritics (tashkeel)	Removes harakat before matching. Recommended ON for most corpora, which are typically unvowelled.
Generate KWIC context files	Produces a KWIC file for each pattern (top 10 words, up to 10 contexts each). Uncheck to skip and reduce processing time on large corpora.

3.2 Right Panel — Output

The right panel displays real-time output across three tabs.

Log Tab

Displays the full processing log with colour-coded message levels: INFO, FILE, CHUNK, OK, WARN, ERROR, PHASE, DONE. The log is saved automatically to session_log.txt at the end of each session.

Results Tab

Displays the frequency table for the selected pattern after analysis completes. Five columns are shown:

Column	Description
Rank	Display order, 1-based, sorted by frequency descending.
Word	Arabic word form, displayed in large gold text with right-to-left alignment.
Count	Raw occurrence count across the entire corpus.

Column	Description
Pct %	Percentage of the pattern's total token count.
Cumulative %	Running total of Pct% — shown in green for ≤ 50%, amber above.

ℹ Sorting is intentionally disabled. Results are pre-ordered by frequency — the most meaningful order for morphological analysis.

The Search field accepts Arabic text to filter the table in real time. Click any row to open a Concordance Dialog for that word (see Section 3.3).

Statistics Tab

Displays a summary of per-pattern statistics after analysis: unique forms, total tokens, and hapax count. The full statistics report is saved to statistics_report.txt.

3.3 Concordance Viewer

Clicking any word in the Results table opens a Concordance Dialog showing every occurrence of that word in the corpus. The dialog is non-modal, meaning multiple words can be open simultaneously for comparison.

Element	Description
Pivot word	Displayed in large gold text in the dialog header.
Context lines	Left context · ❮ pivot word ❯ · right context, 12 tokens each side.
Source file	Filename shown beneath each line (↳ filename.txt).
Search thread	The corpus search runs in the background — the main interface remains responsive.
Context count	Total number of unique occurrences displayed in the top-right corner.

ℹ Context deduplication is applied: lines that share the same six neighbouring tokens are collapsed to avoid repetition from highly redundant files.

4 Output Folder Structure

Each session creates a versioned, timestamped subfolder inside the selected output folder:

```
MIZAN_v2.0_2026-02-24_17-24/
```

```
│
├── Patterns/
│   ├── تفاعل.txt
│   └── استفعال.txt
│
├── KWIC/
│   ├── تفاعل_KWIC.txt
│   └── استفعال_KWIC.txt
│
├── statistics_report.txt
├── session_summary.txt
└── session_log.txt
```

File	Contents
Patterns/<pattern>.txt	Word-frequency TSV file. Five columns plus attribution watermark header.
KWIC/<pattern>_KWIC.txt	Keyword in Context file. Top 10 words × up to 10 contexts each. Generated only when the option is checked.
statistics_report.txt	Full statistical analysis: corpus profile, per-pattern analysis, cross-pattern comparison and overlap.
session_summary.txt	Brief metadata summary: date, corpus path, format, file count, per-pattern totals.
session_log.txt	Complete timestamped processing log, identical to what appears in the Log tab.

4.1 Pattern Files

Every pattern file begins with an attribution header, followed by tab-separated data:

```
# MIZAN v2.0  —  Arabic Morphological Pattern Analyzer
# Author  : Almoataz B. Al-Said  |  Moataz Tools · 2026
# DOI     : https://doi.org/10.1145/3786605
# Pattern : اِسْتِفْعَال
# Unique  : 125  |  Total : 519
#──────────────────────────────────────────────
Rank    Word        Count   Pct%     Cumulative%
1       25.874   25.874      74      استقصاء
2       36.364   10.490      30      استعداد
```

ℹ The Cumulative% column indicates what proportion of the pattern's total frequency is accounted for by the top N words — a standard measure of vocabulary concentration in corpus linguistics.

4.2 KWIC Files

KWIC files provide qualitative context for the top-10 most frequent words of each pattern:

```
═══════════════════════════════════════════════════════════
  MIZAN v2.0  —  KWIC Contexts
  Pattern : إِسْتِفْعَال
  Top 10 words  ×  up to 10 contexts each
═══════════════════════════════════════════════════════════

  [ 1]  استقصاء   (freq=74)
───────────────────────────────────────────────────────────
   1.  ...وتمحيص رأى الآباء...  [استقصاء]  بعد طول...
       ↳ document_01.txt
```

ℹ Context diversity is enforced: each file contributes at most 2 contexts per word, and exact duplicate sentences are skipped. This ensures that the 10 contexts come from different parts of the corpus rather than a single repetitive source.

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5 Statistics Report

statistics_report.txt contains five major sections, generated automatically at the end of every session.

5.1 Corpus Profile

MIZAN performs a full corpus pre-scan before pattern analysis to count actual Arabic tokens and types. This scan produces precise figures rather than byte-based estimates.

Metric	Description
Total Arabic tokens	Running words — actual count from the corpus.
Total unique types	Vocabulary size — distinct word forms.
Type-Token Ratio	Types ÷ Tokens — measures lexical diversity. Larger corpora typically yield lower TTR; compare only between corpora of similar size.
Per-file breakdown	File size (KB), token count, type count, and TTR for each file. Sorted by size, top 10 shown.

5.2 Per-Pattern Analysis

- Frequency counts: unique forms, total tokens, average frequency per form.
- Hapax analysis: hapax legomena count and percentage, dis legomena count, frequency distribution bands (1–5, 6–20, >20).

- Corpus coverage: pattern token count as a percentage of total corpus tokens.
- Estimated precision score with confidence label (VERY HIGH / HIGH / MODERATE / LOW).
- ASCII frequency bar chart of the top-10 most frequent words.

5.3 Cross-Pattern Comparison

A ranked summary table comparing all patterns analysed in the session by productivity (total tokens), estimated precision, corpus coverage, and lexical richness (hapax ratio).

5.4 Cross-Pattern Overlap Analysis

For sessions with two or more patterns, MIZAN computes pairwise word overlap — the set of word forms that appear in the results of more than one pattern.

Sub-section	Contents
Pairwise table	Shared word count, percentage from each pattern, and top-5 examples.
Per-pattern summary	Total overlap %, exclusive %, and an assessment label.
Interpretation guide	Explains expected overlap ranges and when manual review is warranted.

ℹ Some overlap is linguistically expected and correct. A word such as استعمال may appear legitimately in both a Form X verb pattern and its verbal noun pattern. Overlap indicates shared roots across morphological categories, not necessarily an error.

5.5 Interpretation Reference

Metric	Interpretation
Precision ≥ 90%	VERY HIGH — long patterns with many fixed letters; very few false positives.
Precision ≥ 80%	HIGH — reliable; suitable for dictionary building with minimal review.
Precision ≥ 68%	MODERATE — shorter patterns; spot-check a sample of results.
Precision < 68%	LOW — very short patterns (Form I etc.); results require careful filtering.
Hapax < 40%	Excellent — core vocabulary well represented.
Hapax 40–55%	Good — normal for Arabic morphological patterns.
Hapax 55–70%	Moderate — some rare or contextual matches.

Metric	Interpretation
Hapax > 70%	Review — may indicate false positives or a highly specialised corpus.
Overlap < 5%	Excellent — patterns are highly distinct.
Overlap 5–15%	Good — normal for related Arabic patterns.
Overlap 15–30%	Moderate — spot-check shared words.
Overlap > 30%	High — review whether the patterns are structurally distinct.

6 Troubleshooting

Problem	Solution
GUI does not open	Launch exclusively from cmd.exe: python mizan_v20.py
Pattern not added when typing	Type the pattern and press the ＋ Add button or Enter. Patterns must contain Arabic letters only (ف ع ل notation).
No files found in corpus	Verify that the File Format selection matches your corpus files. MIZAN searches all subfolders automatically.
Arabic words not matched	Ensure Strip diacritics is checked for corpora without vowel marks.
KWIC files not generated	Confirm that Generate KWIC context files is checked in Analysis Options.
Concordance shows no results	The corpus path must be set before clicking a word. If the corpus path has changed since analysis, re-select it.
Very slow processing	Install psutil (pip install psutil). Uncheck Generate KWIC if context files are not required.
Encoding errors in output	Use CP1256 for legacy Windows Arabic files; UTF-16 for Microsoft Word exports.
Empty columns in Results tab	Upgrade to v2.0 — earlier versions used a three-column table.

7 Keyboard Shortcuts

Key	Action
F5	Run Analysis
Escape	Stop Analysis
Ctrl+O	Open Corpus Folder
Ctrl+S	Save Log

Key	Action
Ctrl+L	Clear Log
Ctrl+Q	Quit
Enter	Add pattern (when focus is in the pattern input field)

8 Citation

Please include the following citation when using MIZAN in academic publications, theses, or research reports:

Al-Said, A. B. (2026). Mizan: An Innovative Tool for Building Arabic Morphological Pattern Dictionaries and Enhancing Lexical Resource Development. *ACM Transactions on Asian and Low-Resource Language Information Processing*, 25(2), Article 11.

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```

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